

SATYAJIT BARUA

Front-End Developer • React Specialist • QA Automation Engineer

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PROFESSIONAL SUMMARY

Front-End Developer and QA Automation Engineer with 10+ years of experience building and testing enterprise-grade web applications. Proficient in React 18+, TypeScript, and modern front-end ecosystems including Tailwind CSS, Styled Components, and React Router v6. Experienced in custom hooks, Context API, Suspense/lazy loading, and performance optimization. Strong QA background with Azure DevOps, Playwright, and Selenium — uniquely positioned to bridge development and quality assurance in Agile environments.

REACT & FRONT-END SKILLS

React Core	React 18+, JSX, Functional Components, Error Boundaries, React.memo
Hooks	useState, useActionState, useEffect, useContext, useCallback, useMemo, useReducer, useRef
Custom Hooks	useFetch, useForm, useDebounce, useLocalStorage — reusable logic extraction
State Management	Context API, Redux Toolkit, Zustand, React Query / TanStack Query
Routing	React Router v6 — nested routes, protected routes, loaders, lazy routes
Performance	Code splitting, React.lazy, Suspense, memoization, Web Vitals, bundle analysis
Data Fetching	Fetch API, Axios, Promises, async/await, React Query caching & mutations
Forms	Controlled & Uncontrolled components, useRef, React Hook Form, real-time validation
Styling	Tailwind CSS, DaisyUI, Styled Components, CSS Modules, CSS Variables
Tooling	Vite, Next.js, TypeScript, Storybook, ESLint, Prettier, VS Code
Testing	Jest, React Testing Library, Vitest, Playwright v1.42, Selenium WebDriver v4
Accessibility	WCAG 2.1, ARIA roles, 508 compliance, keyboard navigation, JAWS screen reader

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript, HTML5, CSS3, Python, C#, C/C++
Test Management	Microsoft Azure DevOps, HP ALM / HP QC, JIRA, Team Foundation Server (TFS)
Automation	Playwright v1.42, HP Unified Functional Testing (UFT), Selenium WebDriver v4
Databases	MS SQL Server, PostgreSQL, Oracle 11g — DML/DQL query writing
Methodologies	Agile / Scrum, Waterfall, Test-Driven Development (TDD)
OS / Tools	Windows, UNIX/Linux, Git/GitHub, MS Visio, SharePoint, Docker basics

PROJECTS

StudyNook — Study Room Booking Platform [GitHub ↗](#) | [Live Demo ↗](#)

Next.js · React · JavaScript · Tailwind CSS · Zustand · PostgreSQL · Prisma · JWT · Vercel

- Built a full-stack web app where students can list, browse, search, filter, and book study rooms for specific date/time slots.
- Implemented JWT-based authentication with protected API routes using Next.js API Routes and Passport.js.
- Used Zustand for client-side state management and Prisma ORM with PostgreSQL for reliable database operations.
- Styled with Tailwind CSS for a fully responsive layout across all device sizes; deployed to Vercel.

Online Book Borrowing Platform [GitHub ↗](#) | [Live Demo ↗](#)

Next.js · React · JavaScript · Tailwind CSS · MongoDB · Mongoose · JWT · BetterAuth · Vercel

- Developed a digital library platform allowing users to browse a book catalog, filter by category, and borrow titles online.
- Built secure user authentication with JWT and bcryptjs; managed user profiles and borrow history via Next.js API routes.
- Modeled data with Mongoose schemas for User and Book collections in MongoDB Atlas.
- Implemented SSR via Next.js for fast initial page loads and improved SEO indexability.

KeenKeeper — Data Visualization Dashboard [GitHub ↗](#) | [Live Demo ↗](#)

React.js · JavaScript · Tailwind CSS · React Router DOM · Recharts · Vite · Netlify

- Built an interactive data visualization dashboard using Recharts to render charts and graphs from structured data.
- Implemented client-side routing with React Router DOM for smooth multi-page navigation without full page reloads.
- Organized UI into modular, reusable React components with Tailwind CSS for a fully responsive layout.
- Configured Vite for fast development builds and optimized production output; deployed to GitHub Pages and Netlify.

Digital Tools E-Commerce Website [GitHub ↗](#) | [Live Demo ↗](#)

React.js · JavaScript (ES6+) · Tailwind CSS · DaisyUI · React Router DOM · React Toastify · Vite

- Developed a full-featured e-commerce SPA for browsing and purchasing digital tools with product search, filtering, and cart management.
- Used DaisyUI component library on top of Tailwind CSS for a polished, accessible UI with consistent theming.
- Integrated React Toastify for real-time user feedback on cart actions and checkout events.
- Managed persistent cart state using LocalForage; applied React Router DOM for declarative client-side routing.

PROFESSIONAL EXPERIENCE

Front-End Developer & Sr. QA Analyst

Jan 2022 – Present

Caliber Home Loans • Coppell, TX (Remote)

- Built and maintained React 18 component library using functional components, hooks (useState, useEffect, useCallback, useMemo), and Context API — reducing prop-drilling across 10+ pages.
- Created reusable custom hooks (useFetch, useFormValidation, useDebounce) that cut duplicated logic by ~35% across the codebase.
- Implemented React Router v6 with nested routes and route-level code splitting via React.lazy and Suspense, improving initial load time by ~1.4 s.
- Built controlled form components with real-time validation; used uncontrolled inputs with useRef for high-frequency fields to optimize re-render performance.
- Styled UI using Tailwind CSS and DaisyUI alongside Styled Components for theme-aware styles supporting light/dark modes.
- Integrated React Query for server-state management — replacing manual useEffect fetch logic with automatic caching, background refetch, and optimistic updates.
- Optimized script performance via lazy loading, memoized selectors, and bundle analysis — achieving 90+ Lighthouse performance score on key pages.
- Developed automated E2E test scripts using Playwright with TypeScript; implemented page object models, fixtures, and auto-wait mechanisms across Chromium, Firefox, and WebKit.

- Collaborated in Agile/Scrum sprints; maintained Azure DevOps for test cases, defect tracking, and RTM reports.

Stack: React 18, TypeScript, Tailwind CSS, DaisyUI, Styled Components, React Router v6, React Query, Playwright, Azure DevOps, SQL Server, Vite, VS Code, Agile

Software QA Tester (Front-End Focus)

Jul 2018 – Dec 2021

Washington State Dept. of Licensing • Olympia, WA (Remote)

- Analyzed functional requirements and wrote detailed test cases for React-based web applications covering UI behavior, API integration, and database validation.
- Performed 508 accessibility compliance testing using JAWS — documenting issues related to ARIA roles, keyboard navigation, and semantic HTML.
- Automated functional and regression tests using HP UFT, creating shared object repositories, reusable actions, and data-driven test tables.
- Used SQL (DML) queries to cross-verify front-end UI data with database records; managed defects and RTM reports in Azure DevOps and TFS 2017.

Stack: HP UFT 12.4, Azure DevOps 2017, TFS 2017, JAWS, SQL Server 2017, HTML, SharePoint 2017, Agile

Software Test Engineer

Mar 2016 – Jun 2018

North Carolina Dept. of Social Services • Raleigh, NC

- Developed Test Plans, Strategies, and Traceability Matrices for multi-tier web applications, ensuring full requirements coverage.
- Executed Smoke, Functional, Regression, Integration, and E2E testing cycles using HP ALM v12.0 as the test management platform.
- Wrote SQL queries to validate data across Oracle 11g tables; automated scripts with HP UFT 12.2 using VBScript and shared object repositories.

Stack: HP ALM 12.0, HP UFT 12.2, Oracle 11g, SQL Server, Jenkins, SVN, HTML, .NET, Agile

QA Tester

Apr 2015 – Feb 2016

Southwest Airlines Co. • Dallas, TX

- Developed and executed test plans for web and client-server applications using TFS 2010; performed backend validation with SQL queries.
- Tested cross-browser compatibility across IE, Firefox, and Chrome; tracked and reported defects throughout the test cycle.

Stack: TFS 2010, HP UFT 12.2, JIRA 7.8.0, Oracle 11g, HTML, .NET, C#, UNIX, Windows, Agile

EDUCATION

B.Sc. in Computer Science & Engineering (CSE)

University of Science and Technology Chittagong, Bangladesh • Minor: Computer Information Systems (CIS)

Bangladesh